SFU

Chapter 2

Spatial Descriptions and Transformation

Overview

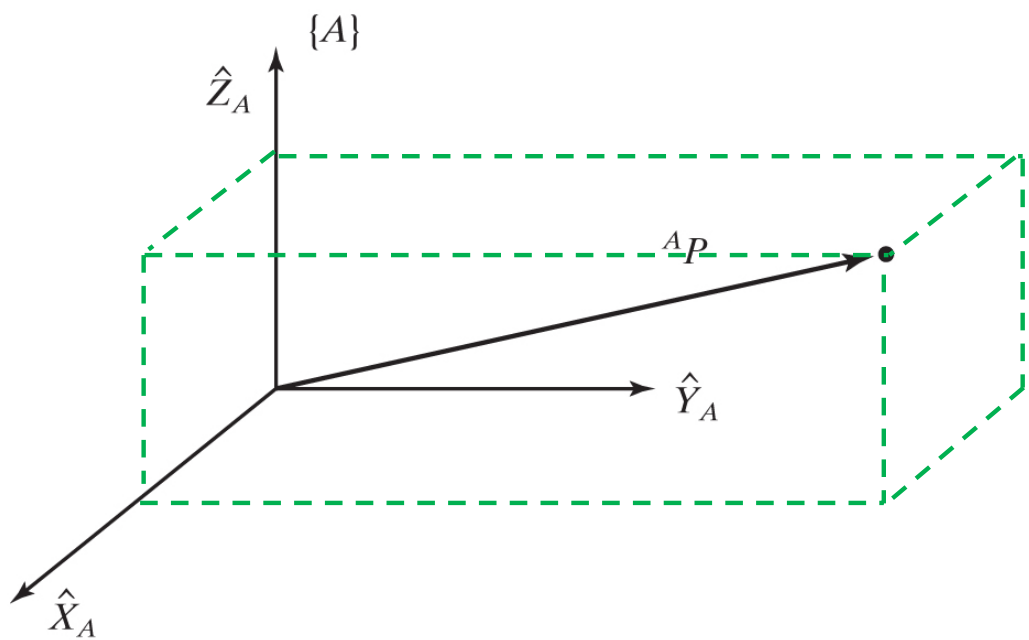
This chapter covers the following topics:

- Descriptions of positions vectors and orientations
- Definition of frames and mapping
- Definition of operators for translations, rotations, and transformations.
- Summary of interpretations
- Transform equations

2.2 Description of a Position

With respect to a reference frame, any point can be described by a 3×1 **position vector**.

$$^{A}P = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$



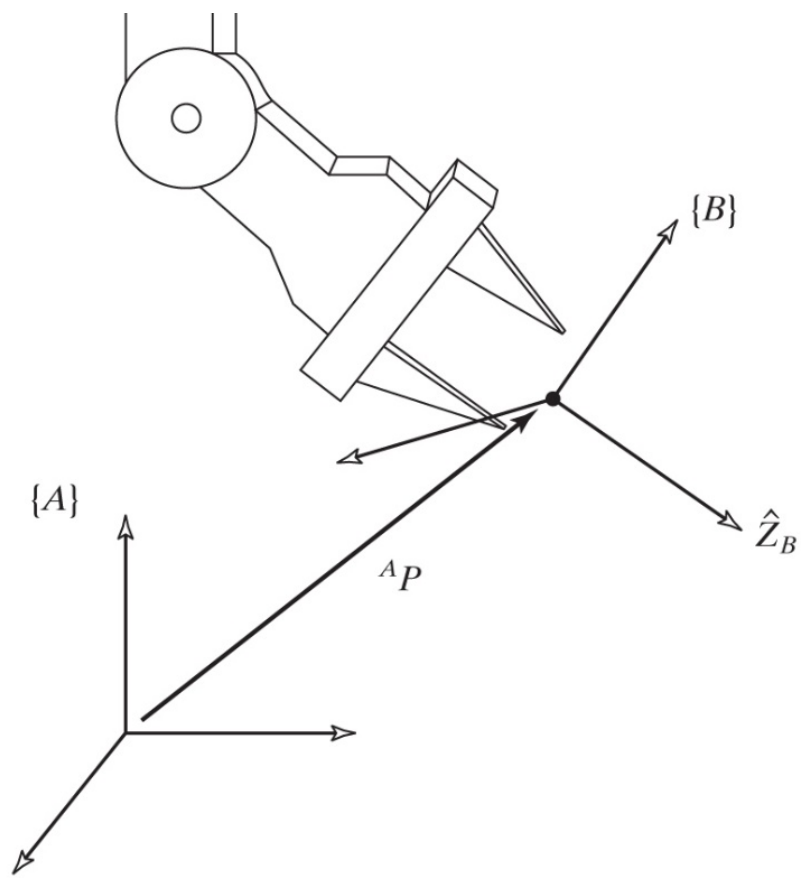
The leading superscript indicates the coordinate system to which the vector is referenced to.

2.2 Description of an Orientation

Rotation matrices are used to describe the projects of the moving frame onto the axes of the fixed frame.

Consider $\{A\}$ be the fixed reference frame and $\{B\}$ be the moving frame with unit vectors $\hat{X}_B$, $\hat{Y}_B$, and $\hat{Z}_B$. Let the unit vectors of frame $\{B\}$ projected on $\{A\}$ become ${}^A\hat{X}_B$, ${}^A\hat{Y}_B$, and ${}^A\hat{Z}_B$.

The relative orientation of $\{B\}$ with respect to $\{A\}$ is defined by a 3×3 orthogonal matrix as



$$ {}^A_BR = \begin{bmatrix} {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

2.2 Description of an Orientation – Direct Cosine Representation

The projection of each axis of moving frame $\{B\}$ onto the axes of the fixed frame $\{A\}$ can be represented by direction cosines.

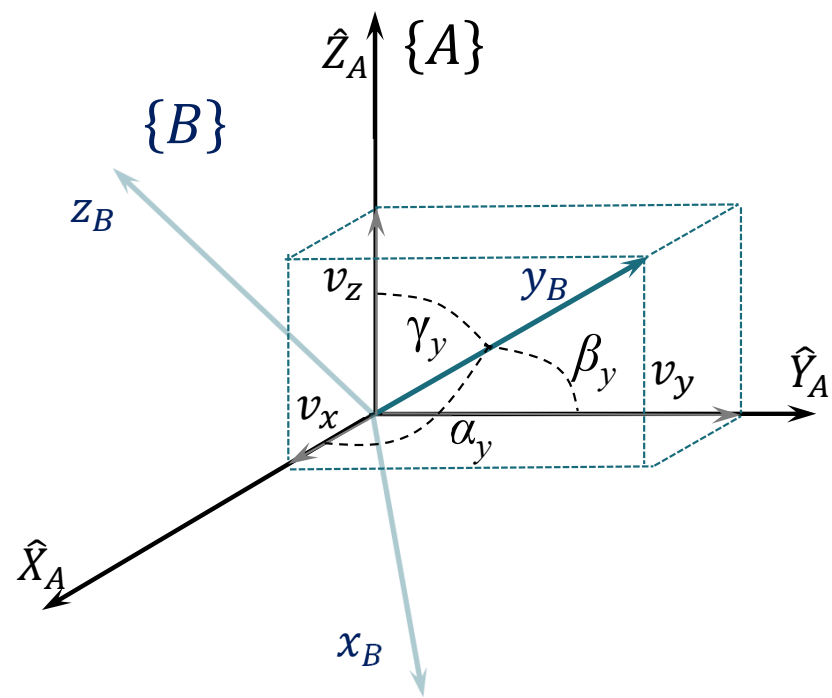
For example, axis y_B projected onto the axes $\hat{X}_A$, $\hat{Y}_A$, and $\hat{Z}_A$ requires knowing the angles α_y, β_y, and γ_y.

$$^{A}\hat{Y}_B = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} \cos(\alpha_y) \\ \cos(\beta_y) \\ \cos(\gamma_y) \end{bmatrix}$$

Since all the axes are unit vectors, the dot product definition reduces to

$$\cos(\alpha_y) = \frac{\hat{X}_A \cdot \hat{X}_B}{\|\hat{X}_A\|\|\hat{Y}_B\|} = \hat{X}_A \cdot \hat{Y}_B$$

where $\|\hat{X}_A\| = \|\hat{Y}_B\| = 1$.



Credit: Dr. F. Firmani

2.2 Description of an Orientation – Direct Cosine Representation

Therefore, the rotation matrix

$$ {}^{A}_{B}R = \left[{}^{A}\hat{X}_B \quad {}^{A}\hat{Y}_B \quad {}^{A}\hat{Z}_B \right] = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix} $$

can be defined in terms of direction cosines as

$$ {}^{A}_{B}R = \begin{bmatrix} cos(\alpha_x) & cos(\alpha_y) & cos(\alpha_z) \\ cos(\beta_x) & cos(\beta_y) & cos(\beta_z) \\ cos(\gamma_x) & cos(\gamma_y) & cos(\gamma_z) \end{bmatrix} $$

or in terms of dot products

$$ {}^{A}_{B}R = \begin{bmatrix} \hat{X}_A \cdot \hat{X}_B & \hat{X}_A \cdot \hat{Y}_B & \hat{X}_A \cdot \hat{Z}_B \\ \hat{Y}_A \cdot \hat{X}_B & \hat{Y}_A \cdot \hat{Y}_B & \hat{Y}_A \cdot \hat{Z}_B \\ \hat{Z}_A \cdot \hat{X}_B & \hat{Z}_A \cdot \hat{Y}_B & \hat{Z}_A \cdot \hat{Z}_B \end{bmatrix} $$

2.2 Description of an Orientation

<u>*Note*</u>: the description of frame $\{A\}$ relative to $\{B\}$ is given by the transpose of ${}^{A}_{B}R$, that is

$$ {}^{B}_{A}R = {}^{A}_{B}R^{T} $$

<u>*Note*</u>:

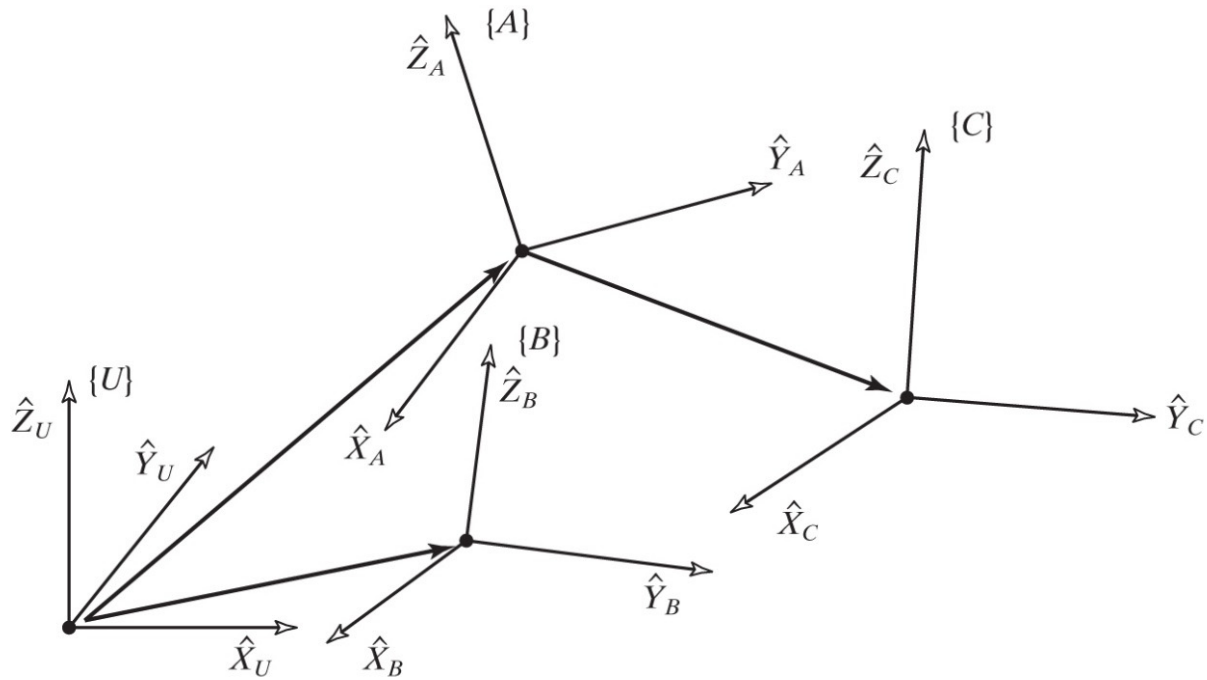
$$ {}^{A}_{B}R^{T}\ {}^{A}_{B}R = \begin{bmatrix} {}^{A}\hat{X}_{B}^{T} \\ {}^{A}\hat{Y}_{B}^{T} \\ {}^{A}\hat{Z}_{B}^{T} \end{bmatrix} \begin{bmatrix} {}^{A}\hat{X}_{B} & {}^{A}\hat{Y}_{B} & {}^{A}\hat{Z}_{B} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 $$

Hence,

$$ {}^{A}_{B}R = {}^{B}_{A}R^{-1} = {}^{B}_{A}R^{T} $$

2.2 Description of a Frame

Frame is a set of four vectors giving position and orientation information. This description can be thought of as a position vector and a rotation matrix.



For example, frame $\{B\}$ is described by ${}^{A}_{B}R$ and ${}^{A}P_{B_{ORG}}$, where ${}^{A}P_{B_{ORG}}$ is the vector that locates the origin of the frame ${}^{A}P_{B_{ORG}}$.

$$\{B\} = \left\{{}^{A}_{B}R, \quad {}^{A}P_{B_{ORG}}\right\}$$

2.3 Mapping

Mapping is used in robotics to change the descriptions of quantities in terms of various reference coordinate systems, form frame to frame.

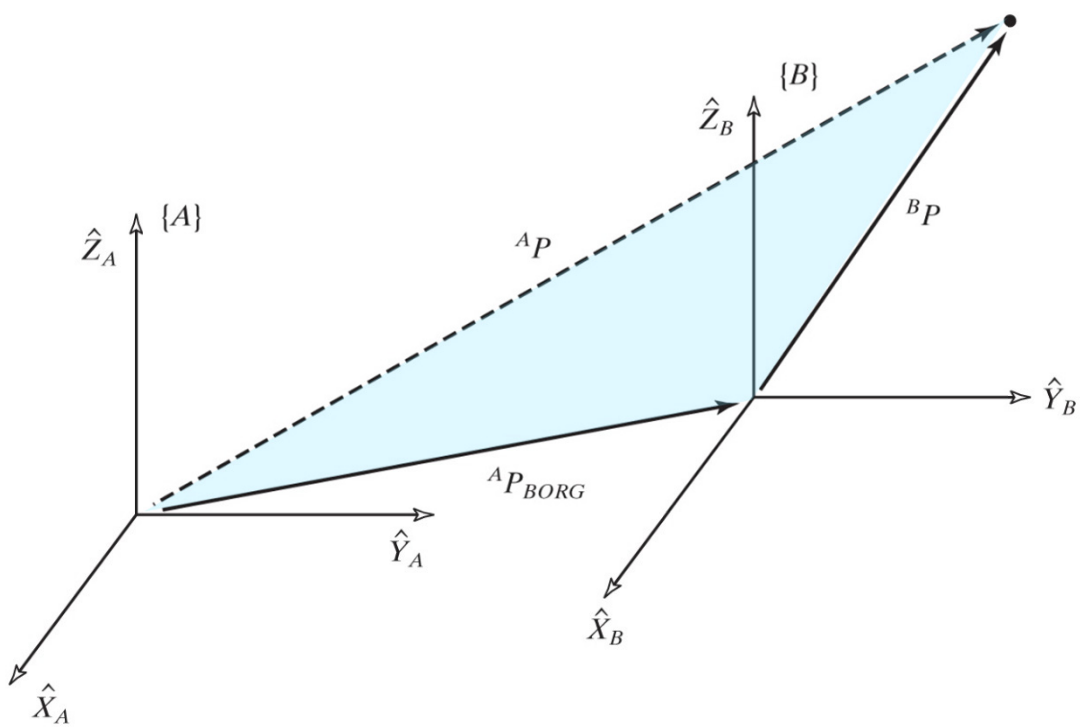
Types of mapping:

- Translational mapping
- Rotational mapping
- General mapping

2.3 Mapping – Translational

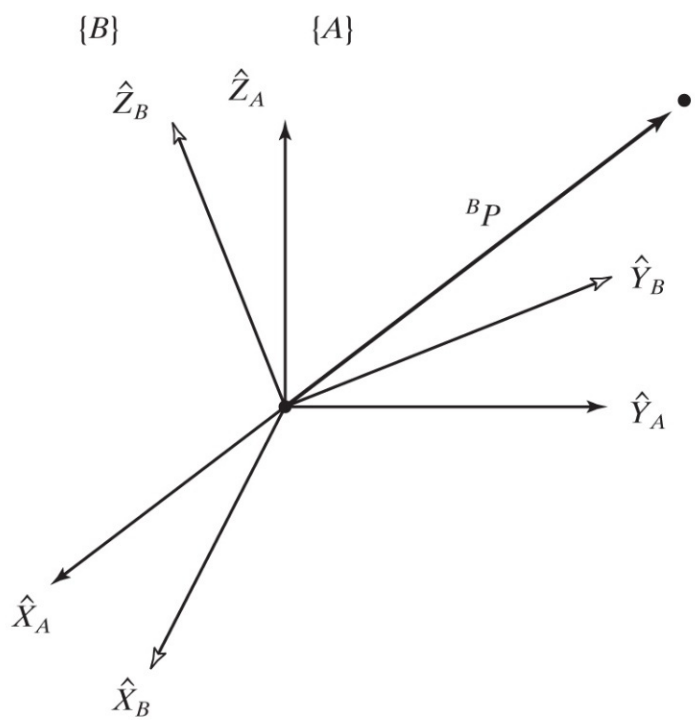
Consider the position defined by the vector ${}^{B}P$ as shown. Let us express it in space in terms of frame $\{A\}$, that is ${}^{A}P$. Note that $\{A\}$ and $\{B\}$ have the same orientation. Given that vector ${}^{A}P_{B_{ORG}}$ locates the origin of $\{B\}$ with respect to $\{A\}$, we have

$$ {}^{A}P = {}^{B}P + {}^{A}P_{B_{ORG}} $$



2.3 Mapping – Rotational

Let the two frames $\{A\}$ and $\{B\}$ share the same origin, but with different orientation. Rotation matrix ${}^{A}_{B}R$ describes $\{B\}$ relative to $\{A\}$.

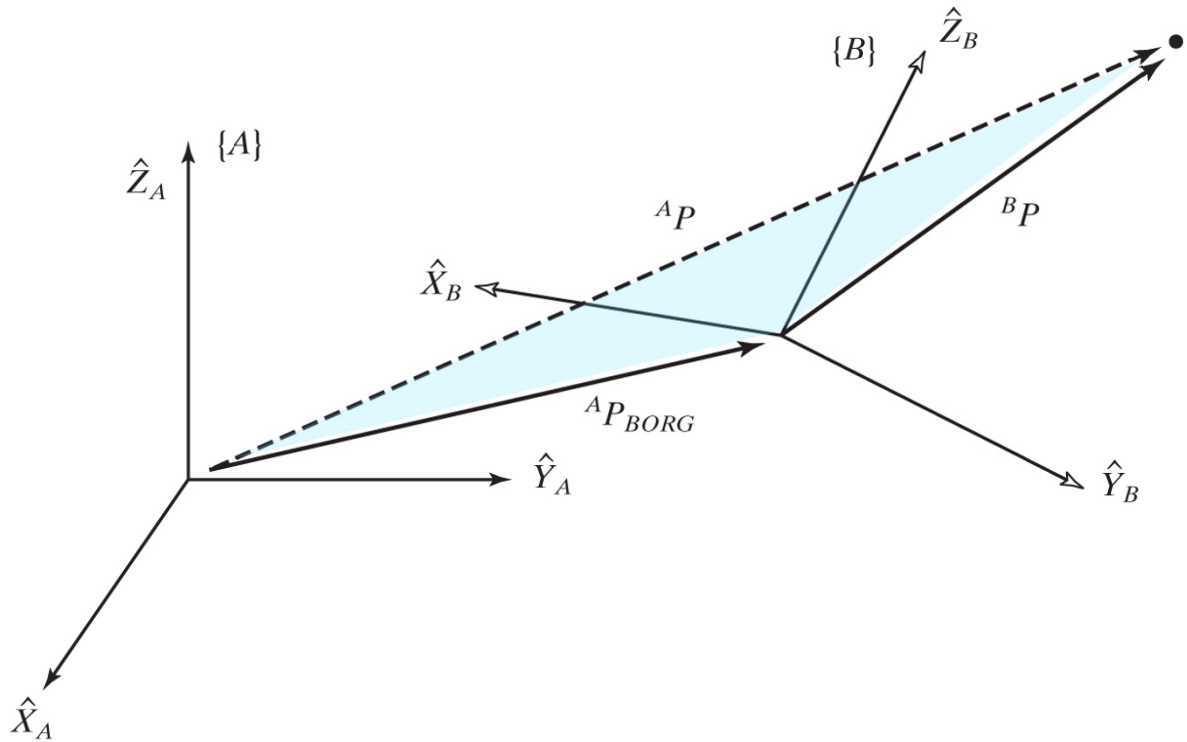


It can be shown that any point expressed in $\{B\}$ can be expressed in $\{A\}$ using the rotational mapping:

$$ {}^{A}P = {}^{A}_{B}R\,{}^{B}P $$

2.3 Mapping – General Mapping

General mapping involves both rotation and translation. Let the two frames $\{A\}$ and $\{B\}$ have different origins and different orientation.



The general transformation mapping of a vector from its description in one frame to a description in a second frame is

$$ {}^{A}P = {}^{A}_{B}R\,{}^{B}P + {}^{A}P_{B_{ORG}} $$

2.3 Mapping – General Mapping

Homogeneous Transform - It is desirable to expressed the general mapping as an operator in matrix form via transformation matrix ${}^{A}_{B}T$.

$$ {}^{A}P = {}^{A}_{B}T\,{}^{B}P $$

Combining both position vector and rotation matrices into one operator:

$$ \begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & {}^{A}_{B}R & & {}^{A}P_{B_{ORG}} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} $$

In which

- a "1" is added s the last element of the 4×1 vector
- a row "[0 0 0 1]" is added as the last row of the 4×4 matrix.

2.4 Mapping with Homogeneous Transforms

Homogeneous transforms can be used for mapping:

i. Translation mapping ($^{A}_{B}R = I_3$)

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \begin{bmatrix} & I_3 & & {}^{A}P_{B_{ORG}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix}$$

ii. Rotational mapping (${}^{A}P_{B_{ORG}} = [0\ 0\ 0\ ]^{\mathrm{T}}$)

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \begin{bmatrix} & & & 0 \\ & {}^{A}_{B}R & & 0 \\ & & & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix}$$

iii. General mapping

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \begin{bmatrix} & {}^{A}_{B}R & & {}^{A}P_{B_{ORG}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix}$$

2.6 Transformation Arithmetic

Homogeneous transforms have the following properties:

i. Compound Transformations

$$ {}^{0}_{n}T = {}^{0}_{1}T\ {}^{1}_{2}T\ {}^{2}_{3}T \cdots {}^{n-2}_{n-1}T\ {}^{n-1}_{n}T = \begin{bmatrix} & {}^{0}_{n}R & & {}^{0}P_{n_{ORG}} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

where ${}^{0}_{n}R = {}^{0}_{1}R\ {}^{1}_{2}R \ldots {}^{n-1}_{n}R$

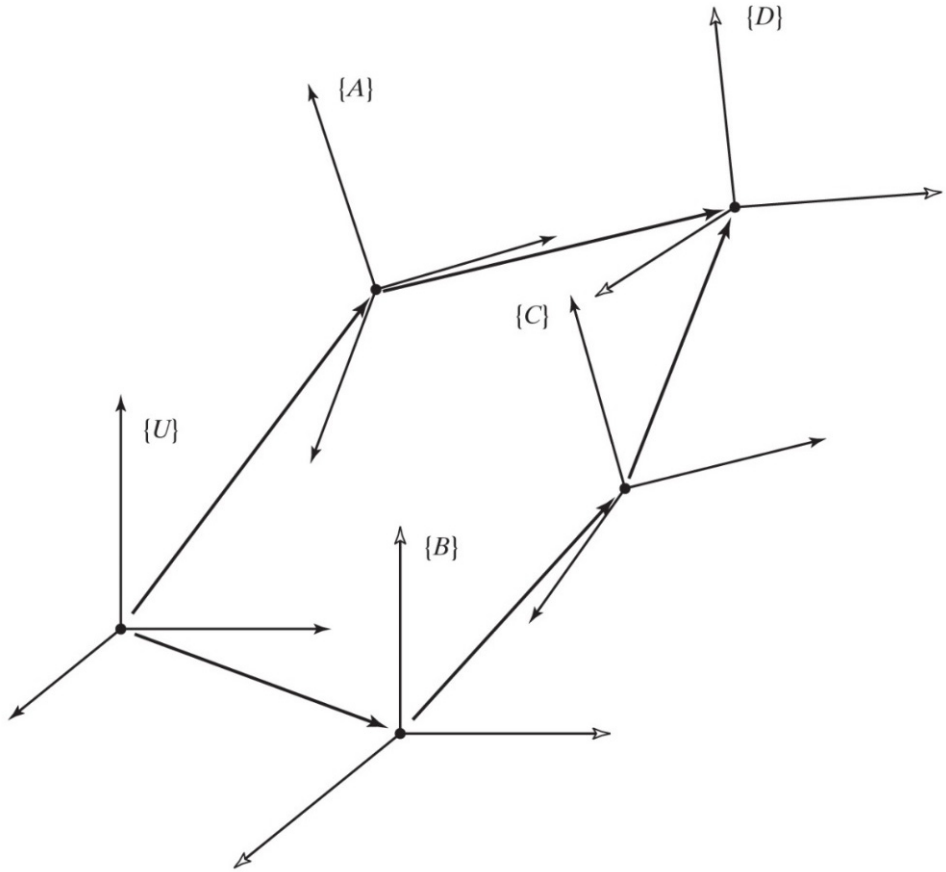
ii. Inversion of a Homogeneous Transform

$$ {}^{A}_{B}T^{-1} = \begin{bmatrix} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T}\ {}^{A}P_{B_{ORG}} \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^{B}_{A}T $$

where

2.7 Transform Equations

Frame $\{D\}$ can be expressed as products of transformations in two different ways.



$$ {}^{U}_{D}T = {}^{U}_{A}T\ {}^{A}_{D}T = {}^{U}_{B}T\ {}^{B}_{C}T\ {}^{C}_{D}T $$

2.8 More on Representation of Orientation – Cayley's Formula

Although a rotation matrix is composed of 9 elements, there are some properties that have to be satisfied, namely orthogonality and unit magnitude, that yield to 6 independent constraints;

$$\underbrace{\begin{aligned} \left\| {}^{A}\hat{X}_B \right\| &= 1 \\ \left\| {}^{A}\hat{Y}_B \right\| &= 1 \\ \left\| {}^{A}\hat{Z}_B \right\| &= 1 \end{aligned}}_{\text{Unit magnitude (3)}} \qquad \underbrace{\begin{aligned} {}^{A}\hat{X}_B \cdot {}^{A}\hat{Y}_B &= 0 \qquad & {}^{A}\hat{X}_B \times {}^{A}\hat{Y}_B &= {}^{A}\hat{Z}_B \\ {}^{A}\hat{Y}_B \cdot {}^{A}\hat{Z}_B &= 0 \qquad & {}^{A}\hat{Y}_B \times {}^{A}\hat{Z}_B &= {}^{A}\hat{X}_B \\ {}^{A}\hat{Z}_B \cdot {}^{A}\hat{X}_B &= 0 \qquad & {}^{A}\hat{Z}_B \times {}^{A}\hat{X}_B &= {}^{A}\hat{Y}_B \end{aligned}}_{\text{Orthogonality (3)}}$$

Thus, a rotation matrix can be represented by only three parameters. Cayley showed that any rotation matrix can be represented with a skew-symmetric matrix S ($S = -S^T$),

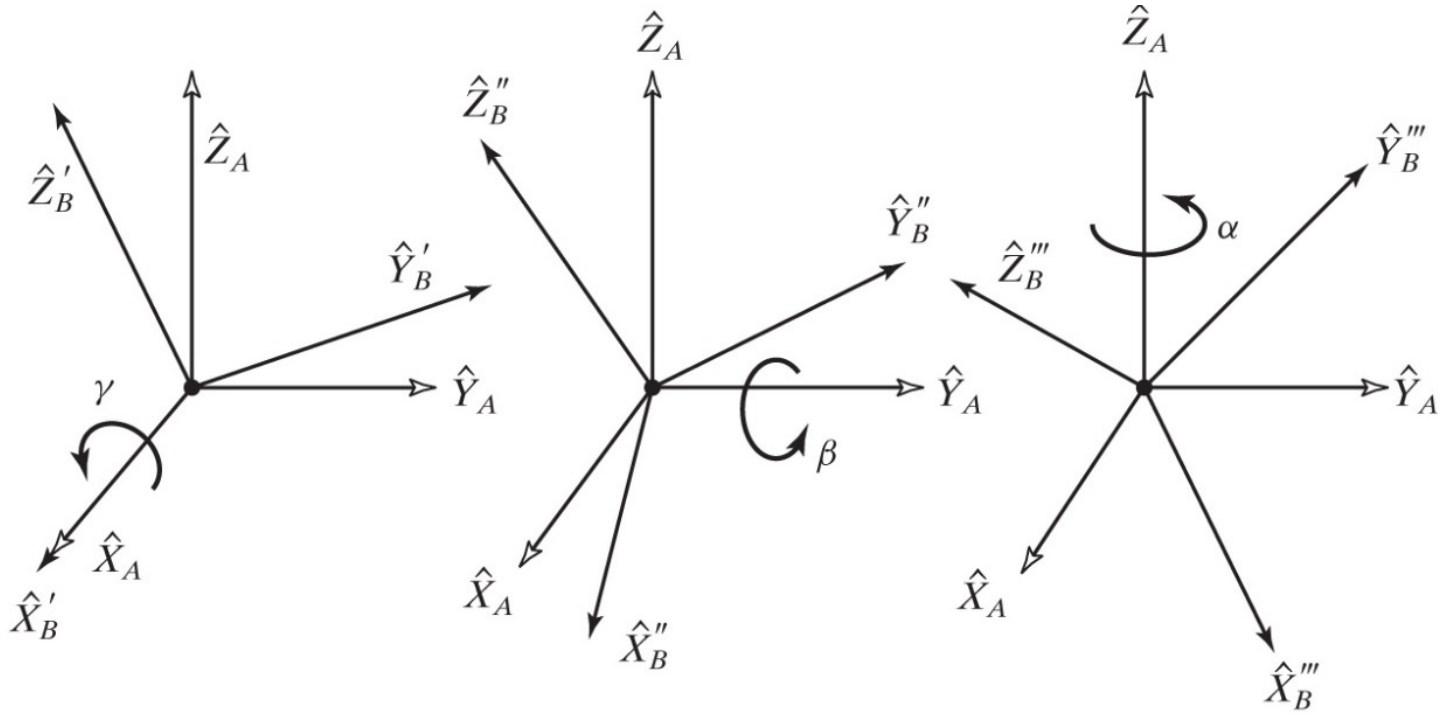
$$R = (I_3 - S)^{-1}(I_3 + S) \qquad \text{where} \qquad S = \begin{bmatrix} 0 & -s_z & s_y \\ s_z & 0 & -s_x \\ -s_y & s_x & 0 \end{bmatrix}$$

2.8 Orientation - Fixed Angles $X - Y - Z$

Let $\{A\}$ be the fixed frame and $\{B\}$ the moving frame. Let them be coincident as well.

"First rotate $\{B\}$ about $\hat{X}_A$ by an angle γ, then rotate $\{B\}$ about $\hat{Y}_A$ by an angle β, and then rotate $\{B\}$ about $\hat{Z}_A$ by an angle α."

Fixed angles are also referred to as **roll**, **pitch**, and **yaw** angles.



Rotations are performed in the order $R_X(\gamma)$, $R_Y(\beta)$, $R_Z(\alpha)$.

2.8 Orientation - Fixed Angles $X-Y-Z$

Forward problem: With the fixed angle representation, the rotation matrices are pre-multiplied.

$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = R_Z(\alpha)R_Y(\beta)R_X(\gamma) = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} $$

$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

Inverse problem: Extracting $X-Y-Z$ angles from the rotation matrix is of interest. Let the rotation matrix be known.

$$ \beta = \mathrm{Atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right) $$

$$ \alpha = \mathrm{Atan2}(r_{21}/c\beta, r_{11}/c\beta\ ) $$

$$ \gamma = \mathrm{Atan2}(r_{32}/c\beta, r_{33}/c\beta\ ) $$

2.8 Orientation - Fixed Angles $X - Y - Z$

<u>*Note*</u>: If $\beta = \pm 90^o$ (so that $c\beta = 0$), the solution of α and γ degenerates. In this case, only the sum or the difference of α and γ can be computed.

Convention:

- If $\beta = 90^o$, then a solution can be calculated to be:

$$\beta = 90^o$$
$$\alpha = 0^o$$
$$\gamma = \mathrm{Atan2}(r_{12}, r_{22})$$

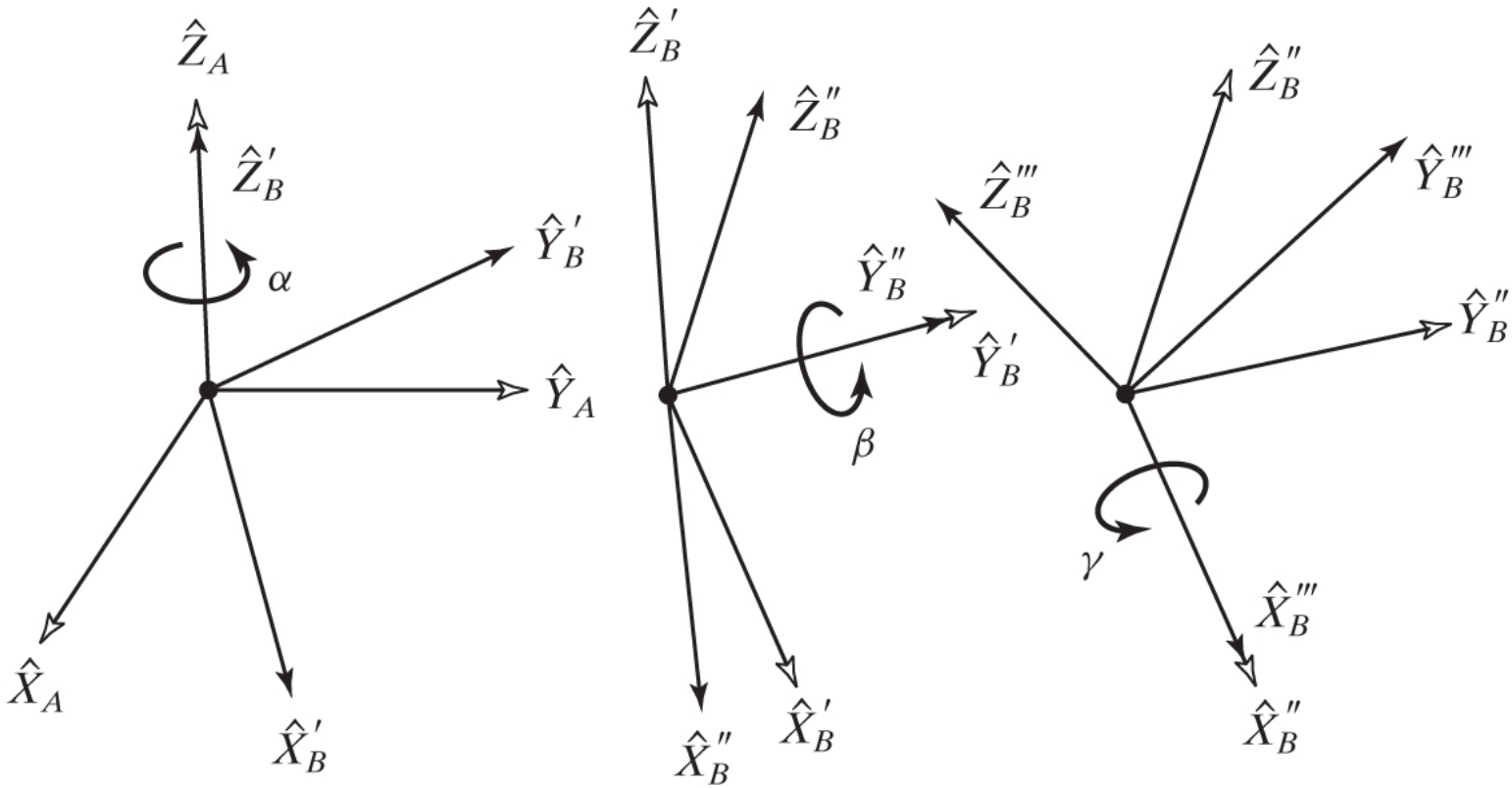
- If $\beta = -90^o$, then a solution can be calculated to be:

$$\beta = -90^o$$
$$\alpha = 0^o$$
$$\gamma = -\mathrm{Atan2}(r_{12}, r_{22})$$

2.8 Orientation - Euler Angles Z – Y – X

Let $\{A\}$ be the fixed frame and $\{B\}$ the moving frame. Let them be coincident.

"First rotate $\{B\}$ about $\hat{Z}_B$ by an angle α, then rotate $\{B\}$ about the rotated $\hat{Y}'_B$ by an angle β, and then rotate $\{B\}$ about the twice-rotated $\hat{X}''_B$ by an angle γ."



A rotation matrix parametrized by Euler angles is indicated by: ${}^A_BR_{Z'Y'X'}(\alpha,\beta,\gamma)$.

Alternative notation: ${}^A_BR_{Z'Y'X'}(\alpha,\beta,\gamma) = {}^A_BR_{ZY'X''}(\alpha,\beta,\gamma)$

2.8 Orientation - Euler Angles Z – Y – X

Forward problem: With the Euler angle representation, the rotation matrices are post-multiplied.

$$
\begin{aligned}
{}^{A}_{B}R_{ZY'X''}(\alpha,\beta,\gamma) &= R_Z(\alpha)R_{Y'}(\beta)R_{X''}(\gamma) \\
&= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}
\end{aligned}
$$

$$
{}^{A}_{B}R_{ZY'X''}(\alpha,\beta,\gamma) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}
$$

Inverse problem: Note that the matrix is exactly the same one as the one obtained for the $X - Y - Z$ fixed angles, but the rotations were taken in opposite order. The equations that solve α, β, and γ are therefore the same, that is

$$
\beta = \mathrm{Atan2}\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right)
$$

$$
\alpha = \mathrm{Atan2}(r_{21}/c\beta, r_{11}/c\beta\ )
$$

$$
\gamma = \mathrm{Atan2}(r_{32}/c\beta, r_{33}/c\beta\ )
$$

2.8 Orientation - Euler Angles Z – Y – Z

"First rotate $\{B\}$ about $\hat{Z}_B$ by an angle α, then rotate $\{B\}$ about the rotated $\hat{Y}'_B$ by an angle β, and then rotate $\{B\}$ about the twice-rotated $\hat{Z}''_B$ by an angle γ."

Forward problem: With the Euler angle representation, the rotation matrices are post-multiplied.

$$
\begin{aligned}
{}^A_B R_{ZY'Z''}(\alpha,\beta,\gamma) &= R_Z(\alpha)R_{Y'}(\beta)R_{Z''}(\gamma) \\
&= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{aligned}
$$

$$
= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}
$$

Inverse problem:

$$
\beta = \mathrm{Atan2}\left(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right)
$$

$$
\alpha = \mathrm{Atan2}(r_{23}/s\beta, r_{13}/s\beta\ )
$$

$$
\gamma = \mathrm{Atan2}(r_{32}/s\beta, -r_{31}/s\beta\ )
$$

2.8 Orientation - Euler Angles $Z - Y - Z$

<u>*Note*</u>: If $\beta = 0^o$ or 180° (so that $c\beta = \pm 1$), the solution of α and γ degenerates. In this case, only the sum or the difference of α and γ can be computed.

Convention:

- If $\beta = 0^o$, then a solution can be calculated to be:

$$\beta = 0^o$$
$$\alpha = 0^o$$
$$\gamma = \text{Atan2}(-r_{12}, r_{11})$$

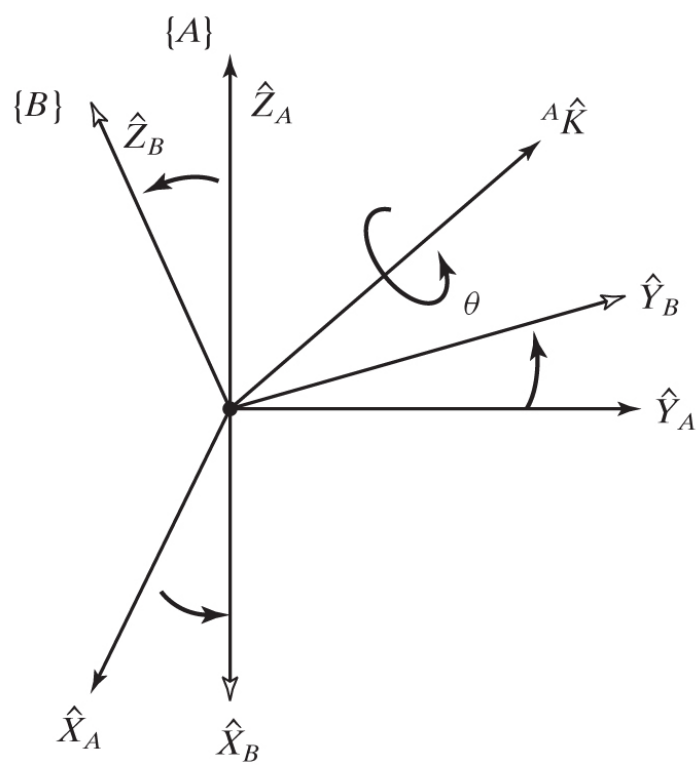
- If $\beta = 180^o$, then a solution can be calculated to be:

$$\beta = 180^o$$
$$\alpha = 0^o$$
$$\gamma = -\text{Atan2}(r_{12}, -r_{11})$$

2.8 Equivalent Angle - Axis Representation

Let $\{A\}$ be the fixed frame and $\{B\}$ the moving frame. Let them be coincident. If the axis of rotation is a general direction (rather than one of the unit directions that we have seen in previous cases), any orientation may be obtained.

"First rotate $\{B\}$ about vector ${}^{A}\hat{K}$ by an angle θ according to the right-hand rule."



A general orientation of $\{B\}$ relative to $\{A\}$ can be written as ${}^{A}_{B}R(\hat{K},\theta)$ or $R_K(\theta)$.

2.8 Equivalent Angle - Axis Representation

Forward problem: for a general rotation, the equivalent rotation matrix is

$$R_K(\theta) = \begin{bmatrix} k_x k_x v\theta + c\theta & k_x k_y v\theta - k_z s\theta & k_x k_z v\theta + k_y s\theta \\ k_x k_y v\theta + k_z s\theta & k_y k_y v\theta + c\theta & k_y k_z v\theta - k_x s\theta \\ k_x k_z v\theta - k_y s\theta & k_y k_z v\theta + k_x s\theta & k_z k_z v\theta + c\theta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

where

$$v\theta = 1 - \cos(\theta) \quad \text{and} \quad {}^{A}\widehat{K} = \left[k_x, k_y, k_z\right]^T$$

Note: the sign of θ is determined by the right-hand rule, with the thumb pointing along the positive sense of ${}^{A}\widehat{K}$.

Inverse problem: for a known rotation matrix, the values of the angle and axis are:

$$\theta = \operatorname{acos}\left(\frac{r_{11} + r_{22} + r_{33} - 1}{2}\right)$$

and

$$\widehat{K} = \frac{1}{2\sin(\theta)} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

2.8 Euler Parameters

Forward problem: Based on the equivalent angle-axis representation $\widehat{K} = [k_x, k_y, k_z]^T$, the Euler parameters are given by

$$\epsilon_1 = k_x \sin\left(\frac{\theta}{2}\right), \qquad \epsilon_2 = k_y \sin\left(\frac{\theta}{2}\right), \qquad \epsilon_3 = k_z \sin\left(\frac{\theta}{2}\right), \qquad \epsilon_4 = \cos\left(\frac{\theta}{2}\right)$$

With the following constraint;

$$\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 = 1$$

The Euler parameters are also known as the **unit quaternion**. The rotation matrix is defined as:

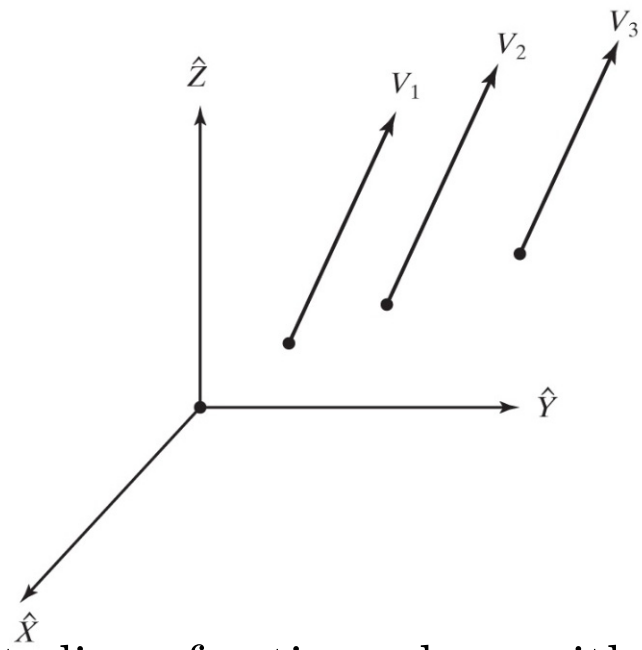
$$R_\epsilon = \begin{bmatrix} 1 - 2\epsilon_2^2 - 2\epsilon_3^2 & 2(\epsilon_1\epsilon_2 - \epsilon_3\epsilon_4) & 2(\epsilon_1\epsilon_3 + \epsilon_2\epsilon_4) \\ 2(\epsilon_1\epsilon_2 + \epsilon_3\epsilon_4) & 1 - 2\epsilon_1^2 - 2\epsilon_3^2 & 2(\epsilon_2\epsilon_3 - \epsilon_1\epsilon_4) \\ 2(\epsilon_1\epsilon_3 - \epsilon_2\epsilon_4) & 2(\epsilon_2\epsilon_3 + \epsilon_1\epsilon_4) & 1 - 2\epsilon_1^2 - 2\epsilon_2^2 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

Inverse problem: For a known rotation matrix, the values of the angle and axis are:

$$\epsilon_1 = \frac{r_{32} - r_{23}}{4\epsilon_4}, \epsilon_2 = \frac{r_{13} - r_{31}}{4\epsilon_4}, \epsilon_3 = \frac{r_{21} - r_{12}}{4\epsilon_4}, \epsilon_4 = \frac{1}{2}\sqrt{1 + r_{11} + r_{22} + r_{33}}$$

2.9 Transformation of Free Vectors

Velocity and force vectors transform differently than the position vectors.



Note: In mechanics, two vectors are equal if they have the same dimensions, magnitude, and direction. Equal vector could have different line of actions.

Line vector refers to a vector that is dependent on its line of action, along with direction and magnitude. Example: Force is a line vector.

Free vector refers to a vector that may be positions anywhere in space without loss or change of meaning, provided that magnitude and direction are preserved. Example: a pure moment vector is a free vector.

Velocity vector written in $\{B\}$, ${}^{B}V$, is written in $\{A\}$ as: ${}^{A}V = {}^{A}_{B}R\ {}^{B}V$

Moment vector written in $\{B\}$, ${}^{B}N$, is written in $\{A\}$ as: ${}^{A}N = {}^{A}_{B}R\ {}^{B}N$

Let The Learning Continue